2022 SSMO Accuracy Round Solutions

SMO Team

AR 1: A bijective function $f : S \to S$, where $S = \{1, 2, 3, 4, 5\}$, is chosen uniformly at random. Find the expected value of $f(1) + f(2) + f(3)$.

Answer: $\boxed{9}$

Solution: We have

$$\mathbb{E}\left(f(1) + f(2) + f(3)\right) = 3 \cdot \mathbb{E}f(1) = 3 \cdot \left(\frac{1+2+3+4+5}{5}\right) = 3 \cdot 3 = \boxed{9}.$$

AR 2: Consider a cone with radius 5 and height 12, and a point P in the same plane as the base of the cone, but a distance of 10 from the center of the base of the cone. We rotate the cone 360° about P such that the plane that the base of the cone lies on stays the same. The volume of the region that the cone sweeps out can be expressed as $m\pi$. Find m.

Answer: $\boxed{1200}$

Solution: Consider the cross section through P and the axis of the cone, placed so that the vertex is at $(10, 12)$ with base endpoints at $(5, 0)$ and $(15, 0)$, and let the point of rotation be $P = (0, 0)$. For a horizontal slice at height x, the outer and inner radii are given by the linear expressions $10 + \frac{5}{12}(12 - x)$ and $10 - \frac{5}{12}(12 - x)$, so the area of the annular cross section is

$$\pi\left[\left(15 - \frac{5}{12}x\right)^2 - \left(5 + \frac{5}{12}x\right)^2\right] = \pi\left(200 - \frac{50}{3}x\right).$$

Thus the volume is

$$\pi\int_0^{12}\left(200 - \frac{50}{3}x\right)dx = \pi\left(200x - \frac{25x^2}{3}\right)\Big|_0^{12} = 1200\pi \implies \boxed{1200}.$$

AR 3: Let $A = (6, 3, 2)$, $B = (2, -9, -6)$, and $O = (0, 0, 0)$. Suppose that D is a point in space such that OD bisects $\angle AOB$ and O, D, A, B are coplanar. In addition, $\angle DAO = 90°$. The length of DO can be expressed as $\frac{a\sqrt{b}}{c}$, where a, b, and c are positive integers, a and c are relatively prime, and b is not divisible by the square of any prime. Find $a + b + c$.

Answer: $\boxed{89}$

Solution: We have

$$AO = \sqrt{6^2 + 3^2 + 2^2} = 7,$$
$$BO = \sqrt{2^2 + (-9)^2 + (-6)^2} = 11, \quad \text{and}$$
$$AB = \sqrt{(6-2)^2 + (3-(-9))^2 + (2-(-6))^2} = \sqrt{224}.$$

Let $\angle AOD = \theta$. By the Law of Cosines on $\triangle AOB$, we have

$$\cos\angle AOB = \cos(2\theta) = \frac{AO^2 + OB^2 - AB^2}{2(AO)(OB)} = \frac{7^2 + 11^2 - 224}{2(7)(11)} = -\frac{27}{77}.$$

Using the half-angle formula,

$$\cos\theta = \sqrt{\frac{1+\cos(2\theta)}{2}} = \sqrt{\frac{1-\frac{27}{77}}{2}} = \frac{5}{\sqrt{77}}.$$

Thus,

$$OD = \frac{AO}{\cos\angle AOD} = \frac{AO}{\cos\theta} = \frac{7}{\frac{5}{\sqrt{77}}} = \frac{7\sqrt{77}}{5} \implies 7 + 77 + 5 = \boxed{89}.$$

AR 4: A monic polynomial f has real roots r, s, t and a monic polynomial g has roots r^3, s^3, t^3. The minimum value of $\frac{g(1)}{f(1)}$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

Answer: $\boxed{91}$

Solution: We have

$$f(x) = (x-r)(x-s)(x-t) \quad \text{and}$$
$$g(x) = (x-r^3)(x-s^3)(x-t^3).$$

So,

$$\begin{aligned}
\frac{g(1)}{f(1)} &= \frac{(1-r^3)(1-s^3)(1-t^3)}{(1-r)(1-s)(1-t)} \\
&= (1+r+r^2)(1+s+s^2)(1+t+t^2) \\
&= \left(\left(r+\frac{1}{2}\right)^2 + \frac{3}{4}\right)\left(\left(s+\frac{1}{2}\right)^2 + \frac{3}{4}\right)\left(\left(t+\frac{1}{2}\right)^2 + \frac{3}{4}\right) \\
&\geq \left(\frac{3}{4}\right)\left(\frac{3}{4}\right)\left(\frac{3}{4}\right) \\
&= \frac{27}{64} \implies 27 + 64 = \boxed{91},
\end{aligned}$$

with equality occurring when $r = s = t = -\frac{1}{2}$.

AR 5: Find the number of ordered pairs (a, b), where $1 \leq a, b \leq 9$, for which the largest integer n that satisfies

$$(a-b)^k \equiv a^k - b^k \pmod{n}$$

for all $k \geq 1$ is $ab - b^2$.

Answer: $\boxed{20}$

Solution: By the Remainder Theorem, $b(a-b) = ab - b^2$ divides $(a-b)^k - (a^k - b^k)$ for every $k \geq 1$. Taking $k = 2, 3$, we get

$$(a-b)^2 - (a^2 - b^2) = -2b(a-b) \quad \text{and} \quad (a-b)^3 - (a^3 - b^3) = -3ab(a-b).$$

Hence, after factoring out $b(a-b)$, the greatest common divisor of the quotients divides $\gcd(2, 3a)$. If a is odd, this gcd is 1, so the largest such n is $ab-b^2$. If a is even, let $c = a-b$. For $k \geq 2$,

$$\frac{(a-b)^k - (a^k - b^k)}{b(a-b)} = -\sum_{j=1}^{k-1} \binom{k}{j} b^{j-1} c^{k-1-j}.$$

Since a is even, b and c have the same parity. If both are odd, the quotient is congruent to $-(2^k - 2) \equiv 0 \pmod 2$. If both are even, every term is even; when $k = 2$, the sole term is $\binom{2}{1} = 2$. Thus, for even a, every difference is divisible by $2b(a-b)$, so the largest such n is not $ab - b^2$. We also need $ab - b^2 > 0$, so $a > b$. For $a \in \{1, 3, 5, 7, 9\}$, there are $a - 1$ choices for b, giving

$$0 + 2 + 4 + 6 + 8 = \boxed{20}.$$

AR 6: Consider an unfair 6-sided die labeled from 1 to 6, such that the probability of rolling a number m is directly proportional to $7 - m$. However, if we roll any number n, then the probability of rolling a number less than n becomes 0, and the probability of rolling any number m from n to 6 inclusive remains directly proportional to $7 - m$. The expected number of rolls until a 6 is rolled can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{197}$

Solution: Let E_x denote the expected number of rolls until a 6 is rolled given that the most recent roll is x. We work our way from E_6 to E_1, which is our answer. Clearly, we have $E_6 = 0$. For E_5, we have

$$\begin{aligned} E_5 &= \frac{2}{3}(E_5 + 1) + \frac{1}{3}(E_6 + 1) \implies \\ E_5 &= \frac{2}{3}E_5 + 1 \implies E_5 = 3. \end{aligned}$$

For E_4 we have

$$\begin{aligned} E_4 &= \frac{3}{6}(E_4 + 1) + \frac{2}{6}(E_5 + 1) + \frac{1}{6}(E_6 + 1) \implies \\ 6E_4 &= 3E_4 + 2E_5 + E_6 + 6 \implies \\ 3E_4 &= 2 \cdot 3 + 0 + 6 = 12 \implies E_4 = 4. \end{aligned}$$

For E_3, we have

$$\begin{aligned} E_3 &= \frac{4}{10}(E_3 + 1) + \frac{3}{10}(E_4 + 1) + \frac{2}{10}(E_5 + 1) + \frac{1}{10}(E_6 + 1) \implies \\ 10E_3 &= 4E_3 + 3E_4 + 2E_5 + E_6 + 10 \implies \\ 6E_3 &= 3 \cdot 4 + 2 \cdot 3 + 0 + 10 = 28 \implies E_3 = \frac{14}{3}. \end{aligned}$$

For E_2, we have

$$\begin{aligned} E_2 &= \frac{5}{15}(E_2 + 1) + \frac{4}{15}(E_3 + 1) + \frac{3}{15}(E_4 + 1) + \frac{2}{15}(E_5 + 1) + \frac{1}{15}(E_6 + 1) \implies \\ 15E_2 &= 5E_2 + 4E_3 + 3E_4 + 2E_5 + E_6 + 15 \implies \\ 10E_2 &= 4 \cdot \frac{14}{3} + 3 \cdot 4 + 2 \cdot 3 + 0 + 15 = \frac{155}{3} \implies E_2 = \frac{31}{6}. \end{aligned}$$

Finally, for E_1, we have

$$\begin{aligned} E_1 &= \frac{6}{21}(E_1+1) + \frac{5}{21}(E_2+1) + \frac{4}{21}(E_3+1) \\ &\quad + \frac{3}{21}(E_4+1) + \frac{2}{21}(E_5+1) + \frac{1}{21}(E_6+1) \implies \\ 21E_1 &= 6E_1 + 5E_2 + 4E_3 + 3E_4 + 2E_5 + E_6 + 21 \implies \\ 15E_1 &= 5\cdot\frac{31}{6} + 4\cdot\frac{14}{3} + 3\cdot 4 + 2\cdot 3 + 0 + 21 = \frac{167}{2} \implies \\ E_1 &= \frac{167}{30} \implies 167+30 = \boxed{197}. \end{aligned}$$

AR 7: After a robber drives in a car for t (not necessarily integral) minutes, the car goes at $120-t$ miles per hour. Whenever the car's speed drops below 60 miles per hour, the robber switches into a new car with no time loss. A police car can drive at a constant speed of 117 miles per hour. Given that the robber starts 1 hour before the police car, how many minutes will pass between when the police car starts and when the police car catches up to the robber?

Answer: $\boxed{216}$

Solution: During each hour, the robber travels at an average speed of 90 miles per hour. Let h be the number of complete hours that have passed for the police car. We have

$$117(h) < 90(h+1) \quad \text{and} \quad 117(h+1) \geq 90(h+2) \implies$$

$$27h < 90 \quad \text{and} \quad 27h \geq 63 \implies h = 3.$$

Suppose the robber is caught t_1 minutes into the fourth hour. After three full hours, the robber has traveled $90\cdot 4 = 360$ miles while the police car has traveled $117\cdot 3 = 351$ miles. We have

$$\frac{120+(120-t_1)}{2}\cdot\frac{t_1}{60} + 360 = 117\cdot\frac{t_1}{60} + 351 \implies$$

$$(240t_1 - t_1^2) + 1080 = 234t_1 \implies t_1^2 - 6t_1 - 1080 = (t_1+30)(t_1-36) = 0 \implies t_1 = 36.$$

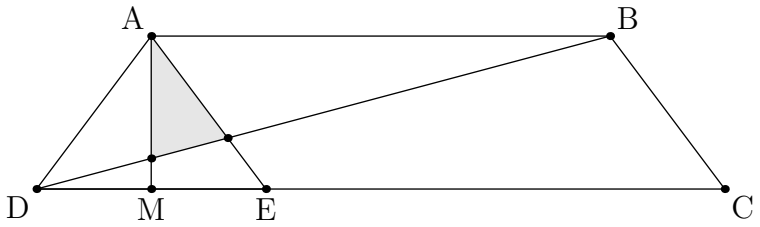
So, the total number of minutes that passed is

$$3\cdot 60 + 36 = 180 + 36 = \boxed{216}.$$

AR 8: Let $ABCD$ be a trapezoid with $AB \parallel CD$. Suppose that $AD = 1$, $DC = 4$, $CB = 2$, and $AB < CD$. Let X be the midpoint of AB and E denote the intersection of AC and BD. If $\angle XEB = \angle ADC$, then AB can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

Answer: $\boxed{15}$

Solution:



Let Y and Z be the feet of the perpendiculars from A and B to CD, respectively. Let $YD = x$. We have

$$BZ = AY = \sqrt{1-x^2} \quad \text{and} \quad ZC = \sqrt{4 - \left(\sqrt{1-x^2}\right)^2} = \sqrt{x^2+3}.$$

From the Law of Sines on triangle XEB, we have

$$\frac{\sin\angle BXE}{BE} = \frac{\sin\angle XEB}{XB} \implies \frac{\sin(\angle XBE + \angle XEB)}{BE} = \frac{\sin\angle ADC}{\frac{AB}{2}}.$$

Note that

$$AB = DC + YD - ZC = 4 + x - \sqrt{x^2+3}$$

and

$$\sin\angle ADC = \sin\angle ADY = \frac{AY}{AD} = \sqrt{1-x^2}.$$

We have

$$\begin{aligned} BE &= \frac{AB}{AB+CD} \cdot BD \\ &= \frac{AB}{AB+CD} \cdot \sqrt{DZ^2 + BZ^2} \\ &= \frac{AB}{AB+CD} \cdot \sqrt{(CD-ZC)^2 + BZ^2} \\ &= \frac{4+x-\sqrt{x^2+3}}{8+x-\sqrt{x^2+3}} \cdot \sqrt{\left(4-\sqrt{x^2+3}\right)^2 + \left(\sqrt{1-x^2}\right)^2} \end{aligned}$$

and

$$\begin{aligned} \sin(\angle XBE + \angle XEB) &= \sin(\angle BDC + \angle ADC) \\ &= \sin\angle BDC \cdot \cos\angle ADC + \cos\angle BDC \cdot \sin\angle ADC \\ &= \frac{BZ}{BD} \cdot \left(-\frac{YD}{AD}\right) + \frac{DZ}{BD} \cdot \frac{AY}{AD} \\ &= \frac{\sqrt{1-x^2} \cdot \left(4-\sqrt{x^2+3}\right) - x\sqrt{1-x^2}}{\sqrt{\left(4-\sqrt{x^2+3}\right)^2 + \left(\sqrt{1-x^2}\right)^2}}. \end{aligned}$$

Substituting back into the Law of Sines relationship on triangle XEB, we have

$$\begin{aligned} \frac{\frac{\sqrt{1-x^2}\cdot\left(4-\sqrt{x^2+3}\right)-x\sqrt{1-x^2}}{\sqrt{\left(4-\sqrt{x^2+3}\right)^2+\left(\sqrt{1-x^2}\right)^2}}}{\frac{4+x-\sqrt{x^2+3}}{8+x-\sqrt{x^2+3}} \cdot \sqrt{\left(4-\sqrt{x^2+3}\right)^2 + \left(\sqrt{1-x^2}\right)^2}} &= \frac{\sqrt{1-x^2}}{\frac{4+x-\sqrt{x^2+3}}{2}} \implies \\ \frac{\left(4-\sqrt{x^2+3}\right) - x}{\frac{1}{8+x-\sqrt{x^2+3}} \cdot \left(\left(4-\sqrt{x^2+3}\right)^2 + \left(\sqrt{1-x^2}\right)^2\right)} &= 2 \implies \\ \left((4-x) - \sqrt{x^2+3}\right)\left((8+x) - \sqrt{x^2+3}\right) &= 2\left(4-\sqrt{x^2+3}\right)^2 + 2(1-x^2) \implies \\ 4\sqrt{x^2+3} &= 4x+5 \implies \\ x &= \frac{23}{40}. \end{aligned}$$

So,

$$AB = 4 + x - \sqrt{x^2+3} = 4 + \frac{23}{40} - \frac{73}{40} = \frac{11}{4} \implies 11 + 4 = \boxed{15}.$$

AR 9: The graph Γ has 52 vertices labeled $A, A', B, B', \ldots, Z, Z'$, such that A is not connected to A', B is not connected to B', and so on. Suppose that all the vertices other than A' have different degrees (number of connections to the vertex). Find the sum of all possible values for the number of edges (connections) in Γ.

Answer: $\boxed{650}$

Solution: Note that the maximum possible degree of any vertex is 50, since each vertex is not adjacent to its partner. The 51 vertices $A, B, B', \ldots, Z, Z'$ must therefore receive the distinct degrees $\{0, 1, \ldots, 50\}$. Suppose first that A has degree 50. Then A is adjacent to every vertex except A', so there is an edge from A to each of $B, B', \ldots, Z, Z'$. Thus every vertex among $B, B', \ldots, Z, Z'$ has degree at least 1, contradicting the fact that one of the 51 vertices must have degree 0. Hence A cannot have degree 50. Therefore one of $B, B', \ldots, Z, Z'$ must have degree 50. Without loss of generality assume $\deg(B) = 50$. Then B is adjacent to every vertex except B', so every vertex other than B' has degree at least 1. Hence B' must be the vertex with degree 0. We repeat this reasoning with the remaining vertices and remaining unused degrees. Each time we place the largest unused degree on A, we get a contradiction. So, it must lie on the other 25 letters. Eventually, we get that the degrees of A and A' must both be 25. The total number of edges must be

$$\frac{(0 + 1 + \cdots + 50) + 25}{2} = \boxed{650}.$$

AR 10: Let

$$S = \sum_{k=5}^{\infty} 2^{-10k} \binom{2k}{10}.$$

Then the value of S can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find the largest positive integer a such that $2^a \mid mn$.

Answer: $\boxed{10}$

Solution: Note that

$$\binom{2k}{10} = [y^{10}]\,(1+y)^{2k}.$$

Substituting,

$$S = [y^{10}] \sum_{k=5}^{\infty} 2^{-10k}(y+1)^{2k} = [y^{10}] \sum_{k=0}^{\infty} 2^{-10k}(y+1)^{2k}.$$

We have

$$\begin{aligned}
\sum_{k=0}^{\infty} 2^{-10k}(y+1)^{2k} &= \frac{1}{1 - 2^{-10}(y+1)^2} \\
&= \frac{1}{2}\left(\frac{1}{1 - \frac{y+1}{32}} + \frac{1}{1 + \frac{y+1}{32}}\right) \\
&= \frac{1}{2}\left(\sum_{k\geq 0}\left(\frac{y+1}{32}\right)^k + \sum_{k\geq 0}\left(-\frac{y+1}{32}\right)^k\right).
\end{aligned}$$

The coefficient of y^{10} is

$$\begin{aligned} S &= \frac{1}{2}\left(\sum \binom{k}{10}\left(\frac{1}{32}\right)^k + \sum \binom{k}{10}\left(-\frac{1}{32}\right)^k\right) \\ &= \frac{1}{2}\left(\frac{\left(\frac{1}{32}\right)^{10}}{\left(1-\frac{1}{32}\right)^{11}} + \frac{\left(-\frac{1}{32}\right)^{10}}{\left(1+\frac{1}{32}\right)^{11}}\right) \\ &= \frac{1}{2}\left(\frac{32}{31^{11}} + \frac{32}{33^{11}}\right) = \frac{16(33^{11}+31^{11})}{(31\cdot 33)^{11}} \end{aligned}$$

We can lift the exponent to find

$$v_2(33^{11}+31^{11}) = v_2(33+31) = 6,$$

giving an answer of

$$v_2(16) + v_2(33^{11}+31^{11}) = 4+6 = \boxed{10}.$$